

ASIF MAHMUD

INDUSTRIAL & PRODUCTION ENGINEERING GRADUATE

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SUMMARY

Industrial & Production Engineering graduate with strong skills in **mechanical design, CAD modeling, product development, and engineering problem-solving**. Proficient in **SolidWorks** with hands-on experience in 3D modeling, assembly design, design modification, and prototyping. Familiar with **GD&T, tolerance analysis, manufacturability, material selection, 3D printing, and simulation**. Brings practical exposure to manufacturing processes and a detail-oriented approach to developing reliable and manufacturable engineering solutions.

EXPERIENCES

- **Freelance CAD Designer (SolidWorks) | CSWA Certified**
 - Worked as a freelance CAD designer, completing multiple product and engineering design projects using SolidWorks.
 - Applied design principles, design for manufacturability (DFM), and basic engineering analysis in project work.
- **Industrial Trainee** Dec, 2024
Bangladesh Industrial Technical Assistance Centre (BITAC)
 - Completed a 7-day industrial training program with hands-on exposure to various manufacturing technologies.
 - Gained practical experience in conventional and modern manufacturing processes, including machining and fabrication.
- **10 Minute School** Jan, 2020 – Apr, 2020
Content Writer
 - Created educational content tailored for students, focusing on clarity and engagement.
 - Designed informative and visually appealing infographics to support learning materials.

EDUCATION

- **BSc in Industrial and Production Engineering** 2022 – 2026
Ahsanullah University of Science and Technology
CGPA 3.19
- **Higher Secondary School Certificate (HSC)** 2020
Uttara High School and College
GPA 5.00
- **Secondary School Certificate (SSC)** 2018
Uttara High School and College
GPA 5.00

TECHNICAL SKILLS

- **CAD and Design** - SOLIDWORKS (CSWA Certified), AutoCAD, Autodesk Inventor (Basic), CATIA V5 (Basic), Siemens NX (Basic Knowledge), SolidWorks Simulation, SolidWorks Plastics
- **CAE & Simulation** - ANSYS Mechanical, Static Structural Analysis, Mesh Convergence Study, Stress Analysis, Deformation Analysis
- **Engineering Standards** - GD&T, DFM, Engineering Drawings, BOM Preparation, Technical Documentation
- **Programming & Data** - Basic Python, Basic SQL, MATLAB (Basic), Microsoft Excel (Advanced)

CERTIFICATIONS

- Certified SOLIDWORKS CAD Design Associate (CSWA)
- Certified SOLIDWORKS Additive Manufacturing Associate
- ISCEA Global Supply Chain Competition *70% scholarship winner*
- Google Project Management Professional Certificate
- IBM Project Management Specialization, IBM, Coursera
- Lean Project Management Yellow Belt (LeanPM-YB), Lean PM Foundation
- Lean Six Sigma White Belt (CSSC-LSSWB), Council for Six Sigma Certification

SOFTWARE TOOLS AND PLATFORMS

SOLIDWORKS, ANSYS Mechanical, AutoCAD, Autodesk Inventor, CATIA V5, Siemens NX, SOLIDWORKS Simulation, SOLIDWORKS Plastics, Python, MATLAB, SQL, Microsoft Office Suite, Google Workspace (G-Suite), Microsoft Power BI

EXTRACURRICULAR ACTIVITIES

❑ Sub-Executive | AUST Robotics Club (AUSTRC)

- Assisted in planning and coordinating robotics workshops, technical seminars, and club events.
- Collaborated with executive members to organize robotics and automation-related activities.
- Supported member engagement initiatives and promoted participation in technical competitions and workshops.
- Strengthened teamwork, leadership, and event management skills through active club operations.

❑ Sub-Executive | AUST Programming & Informatics Club (AUSTPIC)

- Assisted in organizing programming workshops, technical talks, and skill development sessions.
- Coordinated with club members to ensure the successful execution of technical events.
- Contributed to promotional activities and member communication.
- Enhanced organizational, communication, and problem-solving skills through active participation in club management.

ACHIEVEMENTS

- 1st Runner-Up , TechnoCAD – Inter University CAD Contest 2024
- 70% Scholarship Winner , ISCEA Global Supply Chain Competition 2025

PRODUCT DESIGN AND DEVELOPMENT PROJECTS

Portfolio : grabcad.com/asif.mahmud-11

❑ Ergonomic Medical Chair (ENT/Dental) [🏆 Achieved 2nd Place – Product Showcase]

- Designed an ergonomic medical chair for ENT and dental applications, focusing on patient comfort, practitioner accessibility, and structural stability.
- Applied human-centered design principles to improve usability and adjustability.

❑ Ergonomic Stretcher-cum-Wheelchair [🏆 Achieved 3rd Place – Product Showcase]

- Designed a convertible stretcher-wheelchair for efficient patient transportation in healthcare environments.
- Focused on mobility, ease of operation, user safety, and ergonomic handling.

❑ 4-Track Robotic UGV Chassis

- Designed a modular four-track unmanned ground vehicle (UGV) chassis to enhance mobility and stability on rough terrain.
- Developed a structurally robust chassis emphasizing modularity, maintainability, and terrain adaptability.

❑ Hydraulic Double Chamber Power Lifter

- Designed a hydraulic material lifting system with a double-chamber lifting mechanism for improved load-carrying capacity.
- Applied hydraulic transmission principles and structural design considerations to enhance operational reliability.

❑ Ergonomic Workstation

- Designed an ergonomically optimized workstation to improve operator comfort, productivity, and workplace efficiency.
- Incorporated ergonomic principles to minimize operator fatigue and improve accessibility.

❑ Compact Three-Stage Spur Gear Transmission System

- Developed a compact enclosed three-stage spur gear transmission for efficient speed reduction and torque multiplication.
- Designed gear assemblies, shafts, housing, and supporting components considering mechanical performance and manufacturability.

❑ Modular Welding Station

- Designed a modular welding workstation with integrated tool organization and ergonomic accessibility.
- Optimized the workstation layout to improve workflow, operator safety, and production efficiency.

❑ Solar-Integrated Utility Enclosure

- Designed a solar-powered smart utility enclosure for autonomous operation in off-grid environments.
- Integrated photovoltaic system concepts into an industrial enclosure to support sustainable energy applications.

❑ Automatic Eco-Crop Grind Machine

- Designed an automated agricultural grinding machine aimed at improving processing efficiency and reducing manual effort.
- Developed the mechanical structure and functional layout for reliable continuous operation.